

CartandGo Travel System / Booking Engine Requirements

Client checklist and technical requirements for building a real travel booking engine with automated payment and commission handling.

Prepared for: CartandGo Travels Focus: Travel Engine + Payments Version: Developer Intake Draft

This document leaves branding and marketing content aside. It focuses on the accounts, rules, access, and operations needed to build a functional travel engine where payments and commissions can be processed automatically.

1. Project Goal

Move CartandGo from a travel website into a transactional travel system. Customers should be able to search or request travel services, submit required details, pay securely, and allow the business to track bookings, commissions, payouts, and fulfilment.

Core outcomes expected

- Customers can create travel requests or bookings.
- The platform can collect payments online.
- Commission for the client can be calculated and recorded automatically.
- Where supported by the payment gateway, settlement can be split automatically between the client, supplier, agent, or partner.
- Admins can manage bookings, payments, refunds, suppliers, and commission records.

2. Minimum Information The Client Must Provide

AREA	REQUIRED FROM CLIENT	WHY IT IS NEEDED
Services	List of services for MVP: flights, hotels, visa, tours, insurance, study abroad, loans, courses.	Defines what the first version of the engine must support.
Supplier Access	Flight API, hotel API, consolidator account, or manual supplier workflow.	Real inventory and booking confirmation require supplier access.
Payment Gateway	Paystack, Flutterwave, Stripe, Monnify, or other merchant account credentials.	Needed to collect payments and confirm payment status.
Commission Rules	Exact commission formula by service, route, supplier, agent, or booking type.	Needed to calculate client earnings automatically and correctly.
Settlement Rules	Who receives money, when payouts happen, and who bears gateway fees.	Required for payment splits, payout logic, and finance reports.
Operations	Booking statuses, staff roles, approval process, refund process, cancellation process.	Needed for admin workflow and audit trail.
Legal Policies	Terms, privacy policy, refund policy, cancellation policy, visa disclaimer.	Needed before accepting customer data and payments.

3. Payment Gateway Requirements

The client must provide a live business payment account. The gateway should support API payments, webhooks, refunds, and preferably split settlement or subaccounts.

Required credentials and settings

- Public key.
- Secret key.
- Webhook secret or signature verification key.
- Test/sandbox keys.
- Live/production keys when ready.
- Merchant/business ID, if applicable.
- Settlement bank account details.
- Supported currencies and payment methods.

Recommended payment providers

Paystack Flutterwave Stripe Monnify Interswitch

For a Nigeria-focused travel business, Paystack or Flutterwave is usually practical. If the platform needs advanced international marketplace payouts, Stripe Connect can be considered where supported.

4. Automatic Commission Requirements

The commission logic must be defined before development. The system cannot safely automate commissions if the business rule is vague.

Questions the client must answer

- Who earns commission: company, agent, supplier, affiliate, or partner?
- Is commission a percentage or a fixed amount?
- Is commission calculated on gross payment, supplier cost, markup, or profit?
- Does commission differ for flights, hotels, visa services, tours, or insurance?
- Who bears gateway fees: customer, company, supplier, or shared?
- When does commission become final: on payment, booking confirmation, ticket issuance, or travel completion?
- What happens to commission after refunds, chargebacks, cancellations, or failed bookings?
- Can admin users manually override commission?

Example commission rule

ITEM	AMOUNT	NOTES
Customer payment	NGN 1,000,000	Total amount paid by customer.
Supplier cost	NGN 900,000	Amount owed to airline, hotel, or supplier.
Platform markup	NGN 80,000	Business margin retained by CartandGo.
Agent/client commission	NGN 20,000	Commission recorded and paid according to the business rule.

5. Payment Split / Settlement Model

There are two practical models. The client must choose one before implementation.

Model A: Collect first, settle manually or later

- Customer pays into the CartandGo merchant account.
- The system records supplier cost, commission, gateway fee, and net revenue.
- Admin pays supplier, agent, or partner manually or through a later payout process.
- This is easier to launch and gives the business more control over refunds and failed bookings.

Model B: Automatic split settlement

- Customer pays once.
- The payment gateway splits settlement automatically between configured accounts.

- Requires subaccounts, split codes, bank verification, and clear settlement rules.
- Useful when suppliers, agents, or partners should be paid automatically.

Even when automatic split settlement is used, the platform still needs its own internal ledger. The app must record every payment, commission, refund, payout, and booking status for reporting and dispute handling.

6. Supplier / Travel API Requirements

A real travel engine needs access to live inventory or a defined manual fulfilment workflow. The client must provide supplier details and API access where available.

Flights

- Flight API provider or consolidator account.
- Sandbox and live API keys.
- Supported markets, airlines, cabins, routes, currencies, and passenger types.
- Rules for search, fare hold, ticketing, cancellation, changes, refunds, baggage, and seat selection.
- Whether instant ticketing is allowed or staff must approve before ticket issuance.

Hotels

- Hotel supplier/API account.
- Rate rules, cancellation windows, taxes, fees, and room availability rules.
- Voucher/confirmation format.
- Whether payment is prepaid, pay-at-hotel, or manually confirmed.

Visa and service requests

- Destination countries supported.
- Visa categories supported.
- Document checklist per visa type.
- Service fees, embassy fees, appointment fees, insurance fees, and optional add-ons.
- Approval workflow and customer status updates.

7. Customer Data Requirements

SERVICE	CUSTOMER INFORMATION NEEDED
Flights	Full name as on passport, date of birth, gender, nationality, passport number, passport expiry, email, phone, passenger type, route, travel dates, cabin.
Hotels	Guest names, check-in date, check-out date, number of guests, room preference, special requests, contact details.
Visa	Destination, visa type, travel purpose, passport bio-data page, proof of funds, employment/school/business documents, invitation or itinerary, refusal history if any.
Tours	Destination, dates, number of travelers, budget, package type, preferred activities, contact details.

8. Admin Dashboard Requirements

The admin dashboard is required for managing bookings, finance, and fulfilment.

- View all bookings and service requests.
- View payment status and webhook confirmations.
- View commission per booking.
- Approve or reject booking fulfilment.
- Upload tickets, vouchers, invoices, and receipts.
- Change booking status.
- Handle cancellations and refund requests.
- Manage suppliers, agents, and commission rules.

- Export finance reports.

Suggested booking statuses

Pending Awaiting Payment Paid Processing Held Confirmed Ticketed Cancelled Refunded Failed

9. Suggested Database Records

RECORD	PURPOSE
users	Customers, staff, admins, agents, and partners.
bookings	Main booking/order record for flights, hotels, visas, tours, and services.
booking_passengers	Passenger or guest details linked to a booking.
payments	Gateway transactions, references, amount, currency, status, and webhook data.
commission_entries	Calculated commission, owner, basis, rate, status, and reversal information.
payouts	Settlement or payout records to client, agent, supplier, or partner.
refunds	Refund requests, approvals, gateway references, and commission reversals.
suppliers	Flight, hotel, visa, insurance, and tour suppliers.
audit_logs	Important admin actions and financial changes.

10. Security and Compliance Requirements

- Do not store raw card details in the application database.
- Use payment gateway hosted checkout or tokenized payment methods.
- Keep API keys in environment variables, not source code.
- Verify payment webhooks before marking bookings as paid.
- Restrict admin features by role and permission.
- Log financial changes and commission overrides.
- Protect uploaded customer documents.
- Use HTTPS in production.
- Provide privacy policy and consent for collecting travel documents.

11. Environment Variables Needed

- DATABASE_URL
- AUTH_SECRET or equivalent authentication secret
- PAYMENT_PUBLIC_KEY
- PAYMENT_SECRET_KEY
- PAYMENT_WEBHOOK_SECRET
- PAYMENT_SPLIT_CODE if using split settlement
- FLIGHT_API_KEY
- HOTEL_API_KEY
- APP_URL
- EMAIL_PROVIDER_API_KEY

12. Recommended Build Phases

Phase 1: Foundation

- Database and authentication.
- Booking/request records.
- Payment initialization and webhook verification.

- Commission ledger and admin finance view.
- Manual fulfilment workflow for flights, hotels, and visa services.

Phase 2: Real Supplier Integration

- Flight search API integration.
- Offer selection, passenger form, hold/confirm booking workflow.
- Hotel search and booking integration if required.
- Supplier error handling and booking reconciliation.

Phase 3: Automation and Reporting

- Automatic split settlement where supported.
- Refund and commission reversal workflows.
- Agent/supplier payout dashboard.
- Finance exports and booking analytics.
- Email and WhatsApp notification automation.

13. Final Client Checklist

ITEM	PROVIDED?	NOTES
Payment gateway account and test keys	Yes / No	
Webhook access and production domain	Yes / No	
Settlement bank account details	Yes / No	
Subaccounts or split codes, if automatic settlement is required	Yes / No	
Commission formula for each service	Yes / No	
Refund and cancellation policy	Yes / No	
Flight supplier/API access	Yes / No	
Hotel supplier/API access	Yes / No	
Visa/service workflow and document checklist	Yes / No	
Admin roles and staff permissions	Yes / No	
Terms, privacy policy, refund policy, cancellation policy	Yes / No	